Active Polymers Translocate Faster in Confinement

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(Received 20 October 2025; revised 13 March 2026; accepted 2 June 2026; published 28 July 2026)

Living organisms employ diverse strategies to navigate confined environments. Inspired by translocation observations on California blackworms (*Lumbriculus variegatus*), we combine biological experiments and active-polymer simulations to examine how confinement and stiffness govern translocation. Active filaments translocate fastest when the channel width is comparable to their diameter, with escape time determined by propulsion speed, filament length, and channel geometry. In wider channels, activity and flexibility induce reorientation-dominated conformational changes that prolong escape. A single dimensionless ratio linking confinement to stiffness captures the transition from axis-aligned escape with short wall deflections for stiffer filaments to reorientation-controlled motion with bloblike shapes for flexible filaments. These results provide a unified physical framework for active translocation in confinement and suggest design principles for flexible robotic filaments in complex environments.

DOI: 10.1103/ntm6-dz86

Introduction—Translocation of passive and active flexible filaments through confined channels is fundamental across scales [1–3]. Owing to their thermal nature, passive filaments (e.g., DNA, RNA, proteins) typically require external forces or gradients to traverse nanopores and cellular channels. Their dynamics are governed by chain flexibility [1,4,5], polymer-pore interactions [6,7], confinement geometry [8–10], and external driving [11], relevant to DNA sequencing [12], gene delivery, and nanotechnology [13].

In contrast to passive polymers, active filaments consume energy to sustain persistent movement and undergo shape fluctuations driven by internal forces. The coupling of activity and flexibility gives rise to distinctive conformational dynamics [14–20] and collective behaviors [14,21–24]. Examples span cytoskeletal filaments in cells and macroscopic organisms, including earthworms and snakes navigating pores and crevices. Recent work on transport of active agents in dense or confined media has focused on active colloids [25,26] and rodlike bacteria [27]. By contrast, the translocation of flexible active filaments through confinement remains comparatively underexplored [28].

Passive polymers in confinement are well described by Odijk's deflection-segment picture for narrow channels [29] and de Gennes's blob model for wider ones [30]. A recent study shows that such classical scalings can break down for active filaments when the activity-induced persistence length approaches the blob size [3]. Furthermore, simulations and experiments on active-polymer translocation show that activity speeds up transport [31] and modifies how escape time scales with polymer size [32,33], stiffness [34], and conformation [28], producing behaviors absent in confined passive systems. Because propulsion mechanisms differ across self-propelled filaments, no single scaling applies. For snakelike locomotion, anisotropic friction with the substrate can slow motion as the channel narrows [35,36]; in contrast, worms generate motion primarily through internal actuation with limited wall friction [16], so their confined transport warrants separate examination [33,37,38]. This motivates the central question we address here: how do channel width, flexibility, and activity together govern the active-filament transport? Beyond its fundamental interest for confined active matter, this question is relevant to the design of soft robotic filaments for pipe inspection [39], navigation in tortuous tubular environments [40], and subsurface exploration [41,42]. In this Letter, we investigate how channel width controls the translocation of flexible active filaments using California blackworms (*Lumbriculus variegatus*) experiments and Brownian-dynamics simulations of a tangentially propelled polymer model.

Experimental system—California blackworms are aquatic, slender organisms (diameter $D = 0.5 \pm 0.1$ mm, length $L = 50 \pm 10$ mm) that are both flexible and motile, making them a convenient model for active filaments in open-ended channels (Fig. 1, Supplemental Material, Secs. I and

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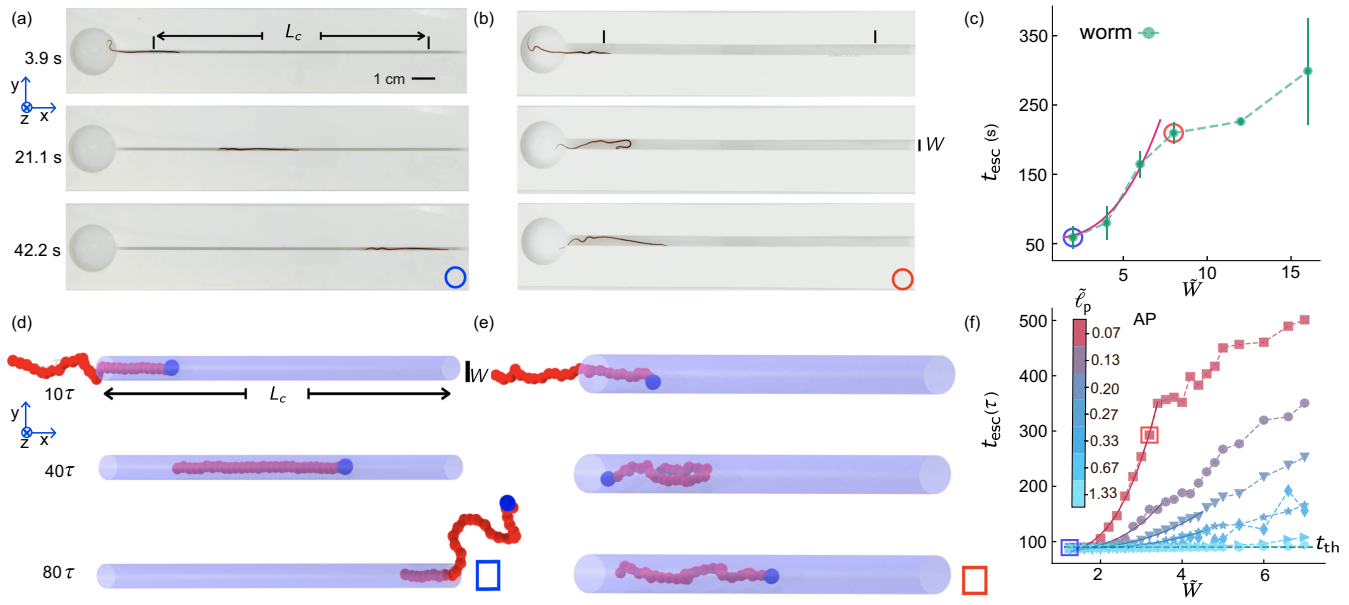


FIG. 1. Translocation and escape time of active filaments in confined channels: (a),(b) Experiments: at scaled width ($\widetilde{W} = 2$), a blackworm escapes within $t_{esc} = 56 \pm 16$ s; in a wider channel ($\widetilde{W} = 8$), it traverses only half the channel over the same interval. (d),(e) Simulation snapshots for active-polymer translocation $\widetilde{\ell_p} = 0.07$: at $\widetilde{W} = 1.4$, strong confinement aligns the filament and speeds escape; in wider channels $\widetilde{W} = 3.0$, frequent reorientations and lateral exploration slow transport. (c),(f) Escape time measurements from both biological experiments and AP simulations showing that the escape time scales with the effective channel width ($2 < \widetilde{W} \leq 8$ for worms and $2 < \widetilde{W} \leq 4$ for AP), $t_{esc} = \widetilde{W}^{\alpha} + C$: worms (solid line) $\alpha = 1.8$; AP simulations $\alpha = [2.8 - 3.4]$ with $C \sim t_{th}$ for $\widetilde{\ell_p} \leq 0.5$. In AP simulations, at higher stiffness, t_{esc} becomes nearly width independent and approaches t_{th}. In wider channels ($\widetilde{W} > 8$ and $\widetilde{W} > 4$ for AP), escape time deviates from the scaling due to the reduced boundary guidance.

II [43]) [16]. The channel connects two reservoirs and is open at both ends; worms can enter and exit from either side, eliminating directional bias.

Active-polymer (AP) simulation—We simulate an active polymer composed of N monomers (diameter σ) with coordinates $\mathbf{r}_i$, propelling along the local tangent, connected by harmonic springs and bending potential, confined within a cylindrical channel. Each monomer i obeys overdamped dynamics, $\gamma \dot{\mathbf{r}}_i = -\nabla_i(U) + \mathbf{F}_i^R + \mathbf{F}_i^A$, where γ is the damping constant and U includes stretching, bending, excluded-volume, and wall-repulsion potentials (Supplemental Material, Sec. III [43]). $\mathbf{F}_i^R$ is the stochastic force with zero mean and variance $2\gamma k_B T$, and the active force on each bead i is $\mathbf{F}_i^A$ with magnitude f_p directed along the tangent, i.e., $\mathbf{F}_i^A = (f_p/2)(\hat{\mathbf{t}}_{i-1,i} + \hat{\mathbf{t}}_{i,i+1})$ for $i = 2, \ldots, N-1$, and $\mathbf{F}_1^A = f_p \hat{\mathbf{t}}_1$, $\mathbf{F}_N^A = f_p \hat{\mathbf{t}}_{N-1}$ for the chain ends. This minimal model serves as a controlled counterpart to the worm experiments by capturing key worm dynamics [33,44,49] and facilitating simulations of self-propelling filaments with tunable motility and flexibility [17,21,50,51].

We systematically vary the channel width W to probe how confinement affects translocation (Fig. 1). The channel length is $L_c = 12$ cm in experiments and $L_c = 60\sigma$ in AP simulations. In AP simulations, we choose ε and σ as energy and length units, respectively, and define the time unit $\tau = \sigma^2\gamma/\varepsilon$ (Supplemental Material, Sec. III [43]). Two dimensionless parameters characterize the problem: the scaled width $\widetilde{W} = W/D$, where D is the filament diameter, indicating available lateral space, and the dimensionless stiffness $\widetilde{\ell_p} = \ell_p/L$, which characterizes the chain flexibility. In simulations, $\ell_p = [\kappa/(k_B T)]$ is the thermal persistence length, where κ denotes the bending rigidity; in experiments, ℓ_p is estimated from tangent-tangent correlations [44]. We also use the Péclet number $\mathrm{Pe} = [(f_p\sigma)/(k_B T)]$, set to 10. We fix $N = 30$ and $k_B T = 0.1\varepsilon$.

Results—We define a successful translocation as an event in which a filament enters the channel from one end and fully exits the opposite end within the observation window. The escape time t_{esc} is measured from head entry to tail exit, capturing the time required for the entire contour to traverse the channel [45] (Supplemental Material, Sec. IV [43]).

Figure 1(a) shows that in narrow channels ($\widetilde{W} \leq 2$; strong confinement) worms align with the channel axis and escape within $t_{esc} = 56 \pm 16$ s, with minimal transverse shape fluctuations. In wider channels ($\widetilde{W} \geq 2$), they adopt undulatory shapes and undergo frequent reorientations, yielding $t_{esc} = 299 \pm 77$ s [5 × longer, Fig. 1(b)]. These observations indicate that stronger confinement enhances directed translocation and shortens escape time.

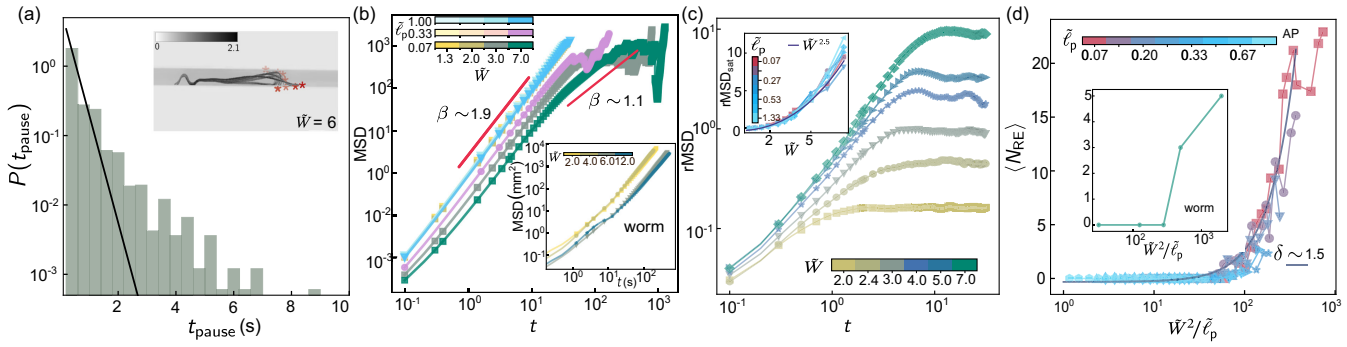


FIG. 2. A single parameter ($\widetilde{W}^2/\widetilde{\ell}_p$) controls active translocation. (a) Pause-duration statistics of the worm when $\widetilde{W} = 6$. The distribution shows an extended tail, indicating intermittent long pauses during the translocation. Inset: overlaid images of the worm during a pause event. (b) Mean-squared displacement (MSD) of the center of mass versus time for varying $\widetilde{W}$ and $\widetilde{\ell}_p$. Decreasing $\widetilde{W}$ increases MSD, indicating enhanced transport in simulations and experiments (inset). (c) Transverse or radial MSD (rMSD), shown for various confinement widths for $\widetilde{\ell}_p = 0.07$, grows then saturates; the saturation level (shown in inset, $\mathrm{rMSD_{sat}}$) increases with $\widetilde{W}$ and is largely stiffness independent, showing that radial exploration is not rate limiting. (d) The average axial reorientation count $\langle N_{\mathrm{RE}} \rangle$, obtained from sign changes of the end-to-end vector $\mathbf{R}_E$, collapses onto a single curve when plotted against $\widetilde{W}^2/\widetilde{\ell}_p$, identifying a single control parameter that links lateral space to stiffness and rationalizes the escape slowdown in wide channels; experiments show similar trends (inset).

AP simulations reproduce these behaviors [Figs. 1(d) and 1(e)]. In narrow channels, $\widetilde{W} \leq 2$, steric interactions align the active filament with the axis, yielding directed motion and faster escape. For wider channels, $\widetilde{W} > 2$, frequent reorientations and conformational fluctuations slow transport, consistent with experiments. In this regime, flexible filaments exhibit bloblike conformations reminiscent of de Gennes–like picture [30] and reorientation-controlled dynamics. Whereas stiffer filaments form short wall-deflection segments and remain axis aligned, reminiscent of an Odijk-like picture [29], which reduces reorientations and favors directed escape.

As shown in Fig. 1(c), worm translocation time increases with the scaled channel width $\widetilde{W}$; worms therefore escape more quickly in narrower channels. In simulations, for ($\widetilde{W} \leq 2$), confinement suppresses transverse fluctuations and the filament adopts a nearly straight conformation, effectively moving as a rigid rod at speed f_p/γ. In this strong-confinement limit, t_{esc} becomes insensitive to flexibility and thermal noise and approaches $t_{\mathrm{th}} = \gamma(L_c + L)/f_p$, where $L = N\sigma$; accordingly, t_{esc} is nearly width independent and close to t_{th} [Fig. 1(f)].

For wider channels ($\widetilde{W} > 2$), both worms and simulations exhibit longer escape times due to increased exploration with frequent reorientations and shape changes [Figs. 1(c) and 1(f)]. In particular, in the intermediate regime ($2 < \widetilde{W} \leq 8$ for worms and $2 < \widetilde{W} \leq 4$ for AP) the data are described by a power-law scaling $t_{\mathrm{esc}} = \widetilde{W}^{\alpha} + C$; for worms $\alpha \sim 1.8$ and in simulations $\alpha \approx 2.8$–3.4 with $C \approx t_{\mathrm{th}}$, reflecting the rigid, propulsion-limited offset.

The difference in scaling exponents reflects two features of the worm experiments that are absent from the constant-propulsion AP model. The first is a transverse exploration space reduced by the quasi-2D experimental geometry, and the second is intermittent stop-go dynamics under confinement [Fig. 2(a) and Supplemental Material, Sec. V [43]]. We define stop-go motion as forward progression interrupted by pauses, during which the center of mass remains nearly stationary while the head explores and reorients locally [Fig. 2(a) inset], and the pause duration t_{pause} measures each such time interval before forward motion resumes. This intermittency is partly intrinsic. Even freely moving worms alternate between crawling, reorientation, and occasional long pauses, with a broader-than-exponential $P(t_{\mathrm{pause}})$ in the absence of confinement (Supplemental Material, Sec. V [43]). Confinement then reshapes this intrinsic intermittency in two ways. Mechanically, it suppresses transverse undulations and the large conformational rearrangements that accompany exploration. Behaviorally, thigmotaxis biases the worm toward forward motion when wall contact is continuous in narrow channels, whereas wider channels permit exploratory wall following that lengthens pauses and broadens $P(t_{\mathrm{pause}})$ [52–54]. In the weak-confinement regime ($\widetilde{W} > 8$ for worms and $\widetilde{W} > 4$ for AP), reduced boundary guidance increases transverse exploration and the escape time deviates from the intermediate-regime power law, through larger reorientations in simulations and longer stop-go pauses in worms (see Supplemental Material, Sec. V [43]). This worm-specific biological layer, absent in the minimal model, contributes to the smaller experimental exponent and the weak-confinement saturation, while the underlying geometric rectification is captured by both systems.

Unlike passive-polymer translocation, where stronger confinement increases entropic resistance and slows

transport [1], prior studies on active polymers in two-dimensional channels [31] and friction-dominated snake locomotion in narrow crevices [35], where stronger confinement slows transport by disrupting the effective driving force or increasing frictional resistance, the present three-dimensional purely steric system is governed by transverse exploration and reorientation dynamics, causing translocation times to increase with channel width. Here, the term "purely steric" refers to the simulation model; real worms may additionally exhibit biomechanical and behavioral responses under confinement.

To characterize filament motion, we measure the mean-squared displacement of the center of mass ($\mathbf{r}_{\mathrm{c.m.}}$) of the polymer, $\mathrm{MSD}(t) = \langle |\mathbf{r}_{\mathrm{c.m.}}(t) - \mathbf{r}_{\mathrm{c.m.}}(0)|^2 \rangle \sim t^{\beta}$. Here, $\langle \cdot \rangle$ denotes averaging over intervals when the filament is fully inside the channel and over successful translocations. Decreasing $\widetilde{W}$ increases the MSD amplitude, indicating enhanced transport for both the AP model and worms [Fig. 2(b), inset for worms]. For ($\widetilde{W} \leq 2$), motion is predominantly deterministic rather than diffusive, with superdiffusive scaling $\beta \sim 1.9$ that is insensitive to stiffness $\widetilde{\ell}_p$. In wider channels, flexible filaments crossover from superdiffusive to near-diffusive dynamics ($\beta \sim 1.1$) as reorientations decorrelate directional persistence, and the MSD saturates at long times due to confinement; stiff filaments remain superdiffusive, consistent with axis-aligned Odijk-like conformations and directed escape. For worms, the dynamics stays superdiffusive while the MSD amplitude decreases in wider channels, in qualitative agreement with the AP model [Fig. 2(b) inset]. A short intermediate subdiffusive plateau appears in worms due to intermittent pauses of the worm, a feature absent in the AP model because propulsion is constant (Supplemental Material, Secs. V and VI [43]).

We next quantify transverse exploration by the radial MSD of monomers, $\mathrm{rMSD} = \langle (1/N) \sum_{i=1}^{N} |\mathbf{r}_i^{\perp}(t+\tau) - \mathbf{r}_i^{\perp}(\tau)|^2 \rangle$, with $\mathbf{r}_i^{\perp}(t) = \{y_i(t), z_i(t)\}$ [Fig. 2(c)]. The rMSD grows and then saturates at a level $\mathrm{rMSD}_{\mathrm{sat}}$ set by the channel width. $\mathrm{rMSD}_{\mathrm{sat}}$ increases with $\widetilde{W}$ and is largely independent of stiffness $\widetilde{\ell}_p$ [inset Fig. 2(c)], with $\mathrm{rMSD}_{\mathrm{sat}} \sim \widetilde{W}^{2.5}$, indicating that self-propulsion enhances transverse fluctuations beyond passive geometric expectations.

Since the saturation time for radial MSD is independent of flexibility (Supplemental Material, Sec. VII, Fig. S8 [43]), delays in translocation are likely set by axial reorientations. We quantify these reorientations from the end-to-end vector $\mathbf{R}_E(t) = \mathbf{R}_N(t) - \mathbf{R}_1(t)$, via its alignment with the channel axis, $\mathrm{sgn}[\hat{\mathbf{R}}_E(t) \cdot \hat{\mathbf{x}}]$. Each sign change marks an axis reversal, and the total count over a trajectory gives N_{RE}.

To rationalize the dependence on confinement and stiffness, we introduce the dimensionless control parameter $\widetilde{W}^2/\widetilde{\ell}_p$, which captures the competition between lateral exploration ($\sim \widetilde{W}^2$) and orientational persistence (set by $\widetilde{\ell}_p$). This scaling arises from the fact that the polymer has roughly L/ℓ_p independent segments, and the reorientation probability in each segment increases with the available lateral space $\widetilde{W}^2$, leading to the control parameter $\widetilde{W}^2/\widetilde{\ell}_p$. The average axial reorientation count per successful translocation $\langle N_{\mathrm{RE}} \rangle$ collapses when plotted against $\widetilde{W}^2/\widetilde{\ell}_p$ [Fig. 2(d)], indicating a single-parameter organization of the dynamics across widths and stiffnesses.

For $\widetilde{W}^2/\widetilde{\ell}_p < 10^2$, $\langle N_{\mathrm{RE}} \rangle \approx 0$, consistent with strong confinement ($\widetilde{W} \leq 2$) or an Odijk-like, axis-aligned regime. As $\widetilde{W}^2/\widetilde{\ell}_p$ increases beyond the threshold, $\langle N_{\mathrm{RE}} \rangle$ rises, signaling a transition to a de Gennes–like regime, where increased lateral space and flexibility promote frequent reorientations and longer residence times. In this regime, $\langle N_{\mathrm{RE}} \rangle \sim (\widetilde{W}^2/\widetilde{\ell}_p)^{\delta}$ with $\delta = [1.5\text{–}1.7]$, corresponding to $\langle N_{\mathrm{RE}} \rangle \sim \widetilde{W}^{[3.0-3.4]}$ at fixed $\widetilde{\ell}_p$. Although the escape time shows a similar growth ($t_{\mathrm{esc}} \sim \widetilde{W}^{[2.8-3.4]}$), many reorientations are small-angle events that do not fully interrupt forward motion. Nevertheless, the observed scaling of t_{esc} and $\langle N_{\mathrm{RE}} \rangle$ indicates that reorientation is the primary source of slowdown for flexible filaments in wider channels. Worms exhibit many reorientations overall but fewer during successful translocations; despite limited statistics, they show the same qualitative trend [Fig. 2(d) inset].

We investigate the role of activity and find that the escape time decreases with activity, approximately as $t_{\mathrm{esc}} \propto 1/Pe$. Although activity increases the free-filament run length ℓ_a, under confinement the translocation dynamics remain governed primarily by wall-mediated interactions and reorientation processes. Accordingly, activity rescales the overall timescale but does not improve the data collapse relative to the confinement-stiffness variable $\widetilde{W}^2/\widetilde{\ell}_p$ (see Supplemental Material, Sec. IX [43]).

In addition to slowing escape, the interplay of filament flexibility and confinement also affects the likelihood of successful translocation. We define the success probability $P_s = N_s/N_t$, where N_s is the number of successful exits and N_t is the number of entries. Failed events ($N_t - N_s$) include exits from the entry side or trapping by excessive reorientations, entanglements, or back-and-forth motion during the observation time. As shown in Fig. 3, the success probability is high ($P_s = 1$) for $\widetilde{W} \leq 2$, independent of filament stiffness $\widetilde{\ell}_p$. However, for $\widetilde{W} > 2.0$, translocation success depends strongly on filament flexibility: Odijk-like, axis-aligned conformations in stiffer filaments yield high success rates ($P_s \gtrsim 0.85$), whereas, flexible filaments in wider channels enter a de Gennes–like regime dominated by reorientations and shape fluctuations, reducing P_s. The dashed boundary $P_s = 0.85$ (green line) separates the Odijk- and de Gennes–like regimes and summarizes the design space for successful translocation.

Outlook—Flexibility, activity, and confinement jointly determine active-filament translocation. Using worms and

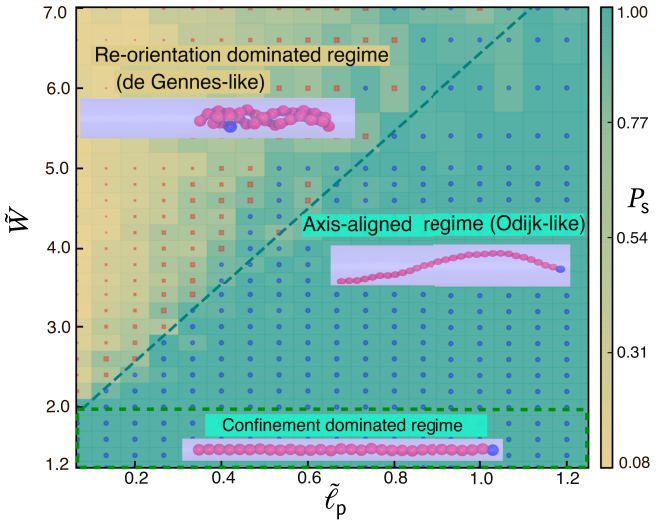


FIG. 3. A simple confinement-stiffness boundary predicts escape success. Translocation success probability P_s versus scaled width $\widetilde{W}$ and stiffness $\widetilde{\ell}_p$. Each symbol represents 200 independent runs, and its area scales with P_s. The dashed line ($P_s = 0.85$) gives an empirical boundary, $\widetilde{W} = 4.83\widetilde{\ell}_p + 1.6$, separating high-success, axis-aligned, Odijk-like transport from low-success, reorientation-dominated, de Gennes–like motion for $\widetilde{W} > 2$. Strong confinement ($\widetilde{W} \leq 2$) yields $P_s \approx 1$. Representative conformations illustrate both regimes.

Brownian-dynamics simulations, we show that narrow channels promote axis alignment and rapid escape, while wider channels permit reorientations and conformational changes that hinder transport. Unlike passive-polymer translocation governed by diffusion and entropic forces, active filaments exhibit confinement- and flexibility-controlled motion that bridges Odijk-like, axis-aligned transport and de Gennes–like reorientation-dominated escape.

Steric confinement biases tangential self-propulsion into axial motion by suppressing orientational fluctuations. In strong confinement, the escape time approaches the intrinsic traversal time t_{th}. In wider channels, transverse exploration and axial reorientations delay escape. The reorientation dynamics are governed by the dimensionless parameter $\widetilde{W}^2/\widetilde{\ell}_p$, capturing the competition between confinement and filament stiffness. The mechanism is distinct from passive-polymer translocation, where narrower channels typically increase the entropic barrier and slow transport [5].

The minimal tangentially propelled filament model used here captures the key ingredients of active translocation and highlights how dimensionality modifies confinement-driven dynamics. Future work should incorporate the intermittent stop-go dynamics of the worm, hydrodynamics, boundary compliance and friction, branched or curved geometries, and multifilament interactions to test the generality of the mechanism and the single-parameter collapse. For a complete picture of pore-to-pore transport in chamber-channel geometries like those in the experiment, filament stiffness affects both channel translocation (the present Letter) and cavity escape [46]; their coupling may yield an optimal stiffness and remains to be explored.

More broadly, these results motivate studies of navigation strategies in flexible organisms, cytoskeletal filaments, and synthetic active matter in complex environments. Finally, the success-probability map and the confinement-stiffness boundary provide physical design rules for autonomous soft robots and microrobot chains navigating confined, irregular pathways [55,56], with applications in pipe inspection [57], drug delivery [58], and medical diagnostics [59].

Acknowledgments—We thank Saeed Mahdisoltani, Harry Tuazon, Arnold Mathijssen, Aparna Bhaskaran, Jennifer Rieser, David Hu, Daehyun Choi, and Kang Sup Lee for helpful discussions. We acknowledge the resources and services provided by the Partnership for an Advanced Computing Environment (PACE) at the Georgia Institute of Technology, U. S. S. B. acknowledges support from NSF Awards No. PHY-2310691 and No. iOS-1941933, National Institutes of Health (NIH) Award No. R35GM142588, and Schmidt Sciences, LLC.

Data availability—The data that support the findings of this article are not publicly available upon publication because it is not technically feasible and/or the cost of preparing, depositing, and hosting the data would be prohibitive within the terms of this research project. The data are available from the authors upon reasonable request.

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Active polymers translocate faster in confinement

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In this supplement, we detail the experimental setup, image-analysis procedures, AP simulation model and parameters, translocation-time definitions, and additional analyses (MSD exponent, radial-MSD saturation, and reorientation metrics). We also provide representative movies from experiments and simulations that support the main results.

I. EXPERIMENTAL SETUP

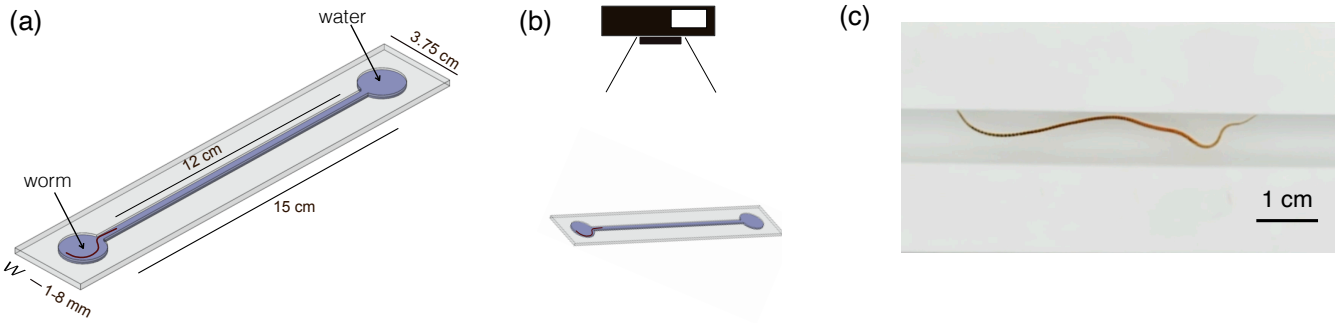


FIG. S1. **Schematic of experimental setup.** (a) Quasi-2D setup used for worm translocation experiments: two circular chambers connected by an open-ended, narrow channel (dimensions annotated in schematic). (b) Imaging setup: a top-mounted camera records motion of the worm between chambers under fixed illumination. (c) Image of California blackworm (*Lumbriculus variegatus*) inside the channel.

California blackworms (*Lumbriculus variegatus*) are freshwater annelids that inhabit lake and river sediments (Fig. S1c, blackworm in a channel). We obtained worms from Carolina Biological Supply and Eastern Aquatics and maintained them in plastic containers ($35 \times 20 \times 12$ cm) with filtered, treated water and stored in a refrigerator at $4°C$, following established protocols [1]. Worms were fed tropical fish flakes once per week, water was refreshed daily. Since blackworms are exempt from institutional animal care review, no additional approval was required. Before experiments, worms were habituated for at least one hour in room-temperature water ($\sim 21°C$) under ambient lighting.

The quasi-2D experimental setup (Fig. S1a) was fabricated from 3 mm thick acrylic sheets, laser-cut (using Trotec Speedy 100 laser cutter) and then bonded together to form two circular chambers connected by a straight channel. Both channel and chambers are open at the top. We fixed the channel length at $L_c = 12$ cm, and varied the channel width W, as given in Table I. For all our experiments, we selected worms with a body length of $L = 5 \pm 1$ cm and diameter $D = 0.5 \pm 0.1$ mm. The persistence length (ℓ_p) of the worm, obtained from tangent-tangent correlations, is 6.5 mm [2].

Channels were filled with filtered, treated water and a single worm was introduced. Experiments were recorded at 30 fps for 1-2 hours using a top-mounted Logitech C920x (Logitech, Taiwan, China) webcam in a light box, without any external perturbations at room temperature (Fig. S1b).

W (mm)	$\tilde{W}$
1	2
2	4
3	6
4	8
6	12
8	16

TABLE I. Channel width W and corresponding scaled width $\tilde{W}$ used in worm experiments.

II. DATA ANALYSIS

We quantified three observables from videos: the translocation (escape) time of the filament, the center of mass of the filament, and the end-to-end distance. Filaments were detected using standard thresholding and contour detection algorithms from the Python OpenCV library (https://github.com/opencv/opencv-python). The raw images were first converted to grayscale and smoothed to reduce noise, after which a binary mask of the filament was obtained through adaptive thresholding. Contours of the segmented filament were then extracted frame-by-frame. These contours were further processed using a skeletonization algorithm from the Python scikit-image library (https://github.com/scikit-image/scikit-image) to compute geometric properties. Although the worm exhibits 3D motion, the setup constrains vertical excursions; we therefore analyze $\{x, y\}$ coordinates only. This approximation does not affect the escape time or center-of-mass dynamics because the out-of-plane fluctuations are suppressed.

III. SIMULATION DETAILS

Active Polymer (AP) model. We consider an active polymer made of N monomers with coordinates r_i, connected by harmonic springs with a bending potential, confined within a cylindrical channel of length L_c and diameter W (Fig. S2a) [3]. The equation of motion for each monomer, i, is

$$\gamma \dot{\mathbf{r}}_i = -\nabla_i (U^{WCA} + U^{Wall} + U^{stretch} + U^{bend}) + \mathbf{F}_i^R + \mathbf{F}_i^A \tag{1}$$

The steric interaction U^{WCA} between all the monomers in the system is modeled via the Weeks-Chandler-Anderson potential [4],

$$U^{WCA}(r) = 4\varepsilon \left[\left(\frac{\sigma}{r}\right)^{12} - \left(\frac{\sigma}{r}\right)^{6} + \frac{1}{4} \right], \tag{2}$$

which vanishes for $r > 2^{1/6}\sigma$. Here ε measures the strength of the repulsive interaction, σ, the size of the particles, and both are set to 1. We measure length in units of particle size σ, energies in ε, and units of time in $\tau = \sigma^2\gamma/\varepsilon$.

We consider an open-ended channel of length L_c and diameter W, Fig.S2(b). The confining wall interaction for each monomer is given by,

$$U^{Wall}(\lambda - |r|) = 4\varepsilon \left[\left(\frac{\sigma}{\lambda - |r|}\right)^{12} - \left(\frac{\sigma}{\lambda - |r|}\right)^{6} + \frac{1}{4} \right], \tag{3}$$

where $\lambda - |r|$ represents the separation distance between the monomer and the wall extending up to $(\lambda - |r| = 2^{\frac{1}{6}}\sigma)$. After accounting for the finite size of the monomer, the available space for the monomer is defined as the effective width $W = 2(\lambda - \frac{\sigma}{2})$, corresponding to the region where the wall repulsion becomes physically significant.

The bonded monomers of the polymer interact via a harmonic potential,

$$U^{stretch}(r) = \frac{k_b}{2}\big(r - R_0\big)^2. \tag{4}$$

Here $R_0 = 1.0\sigma$ is the equilibrium bond length, and $k_b = 4000\ k_BT/\sigma^2$ is the bond stiffness. These parameters make the chain effectively inextensible, diameter of the polymer $D = 1\sigma$ and polymer length $L = N\sigma$.

Any connected triplet in the polymer experiences a bending interaction, which is given by the potential,

$$U^{bend} = \frac{\kappa}{2}\big(\theta - \theta_0\big)^2 \tag{5}$$

Here κ is the bending stiffness and θ is the angle between any consecutive bond vectors and θ_0 is set to π. We vary κ to change to flexibility of the polymer.

$\mathbf{F}_i^R$ represents the stochastic force with zero mean and variance $2\gamma k_BT$ and we kept $k_BT = 0.1\varepsilon$.

In order to mimic the overall behavior of the worm we use a tangential active polymer model, where each bead is self-propelled along the local tangent of the polymer backbone [3, 5, 6]. The last term represents self-propulsion force for the polymer, $\mathbf{F}_i^A = \frac{f_p}{2}(\hat{\mathbf{t}}_{i-1,i} + \hat{\mathbf{t}}_{i,i+1})$ Here, f_p is the strength of the force and $\hat{\mathbf{t}}_{i,i+1} = \mathbf{r}_{i,i+1}/\mathbf{r}_{i,i+1}$ is the unit

tangent vector along the bond connecting beads i and i + 1. This is experienced by the beads ($i = 2, 3, 4...N-1$) of the polymer. The active forces for the end beads ($i = 1$ and $i = N$), as they have only one nearest neighbor, are

$$\mathbf{F}_1^A = \hat{\mathbf{t}}_1 f_p \;\&\; \mathbf{F}_N^A = \hat{\mathbf{t}}_{N-1} f_p \tag{6}$$

We set the number of monomers to $N = 30$ and the channel length to $L_c = 60$. We vary the dimensionless diameter $\tilde{W}$ in the range $[1.245 - 7.000]$ and the dimensionless persistence length $\tilde{\ell}_\mathrm{p} = \ell_\mathrm{p}/L$ in the range $[0.067 - 1.500]$, which control geometric confinement and filament flexibility, respectively. The equations of motion in Eq 1 were integrated with time step, $\delta t = 0.0001\tau$.

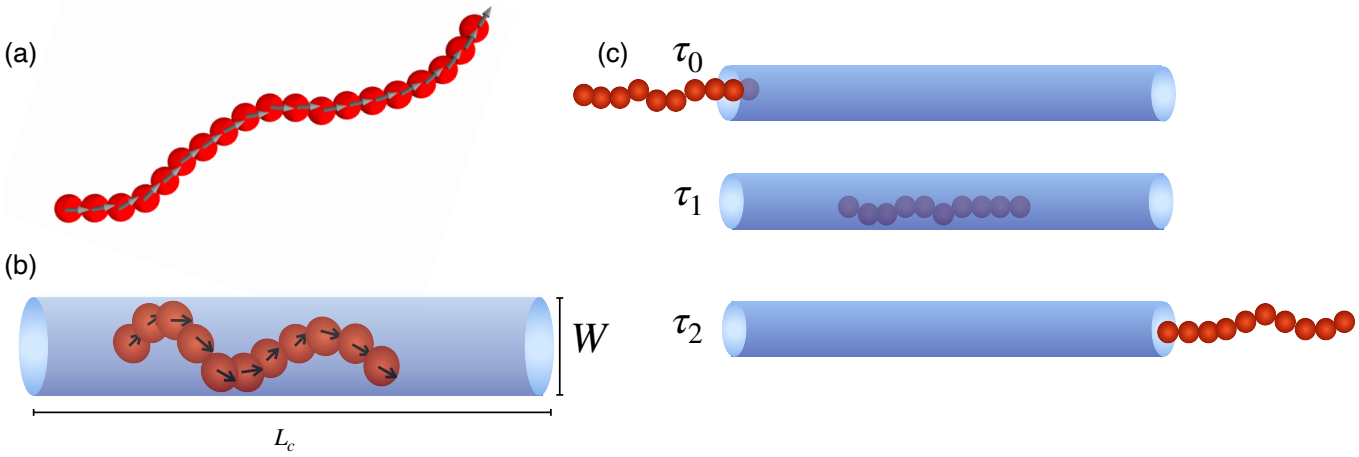


FIG. S2. **Active Polymer (AP) model.** (a) Tangentially-propelled filament: each monomer propels with an active force f_p directed along the local tangent. (b) The active filament moves within an open-ended cylindrical channel of width W and length L_c. (c) Escape time measurement: t_esc is measured from τ_0 (head entry, first monomer) to τ_2 (tail exit, last monomer), including the residence interval τ_1 during which the filament undergoes propulsion-driven motion, reorientations, and conformational fluctuations.

IV. TRANSLOCATION TIME t_esc ANALYSIS

We define a successful translocation as an event in which a filament enters the channel from one end and fully exits the opposite end within the observation window. The translocation time or escape time t_esc is measured from head entry of the filament to tail exit, i.e., $t_\mathrm{esc} = \tau_2 - \tau_0$ (Fig.S2c), naturally capturing the time for the entire contour to traverse the channel. This definition follows established practice for driven translocation [7].

V. STOP-GO DYNAMICS OF THE WORM

In confinement, worms exhibit stop-go dynamics, arising from interactions between the intrinsic locomotion of the worm (worm gait) and the channel boundaries; see Fig.S3. We define stop-go motion as forward progression interrupted by intermittent pauses, during which the center of mass remains nearly stationary while the head undergoes local exploratory and reorientation motion (Fig.S3). Experimentally, pause durations are identified from the center-of-mass trajectory, x_{cm}, as consecutive intervals satisfying $\frac{\Delta x_{cm}}{\Delta t} \leq 10^{-4}$, where Δt is the time between video frames.

In the intermediate-confinement regime ($2 < \tilde{W} < 8$ for worms and $2 < \tilde{W} < 4$ for AP model), boundary encounters are frequent and continue to guide the filament. These wall contacts suppress large deformations and large-angle reorientation events, so the escape time grows approximately as a power law with $\tilde{W}$. In worms, this regime is associated primarily with short pauses (Fig.S4(a,d)), although longer pause events begin to appear as $\tilde{W}$ increases (Fig. S4 (b,e)). Repeated boundary encounters nevertheless maintain overall directed motion. The power-law fit in Fig. 1c is therefore intended for this intermediate-confinement regime.

At larger widths ($\tilde{W} > 8$ for worms), boundary guidance is reduced, the worm can deform more freely, and stop-go dynamics becomes more prominent. This is evident from the broader pause-time $P(t_\mathrm{pause})$ distribution and the longer vertical segments in the kymographs (Fig.S4(c,f)). In this weak-confinement regime, a substantial fraction of the

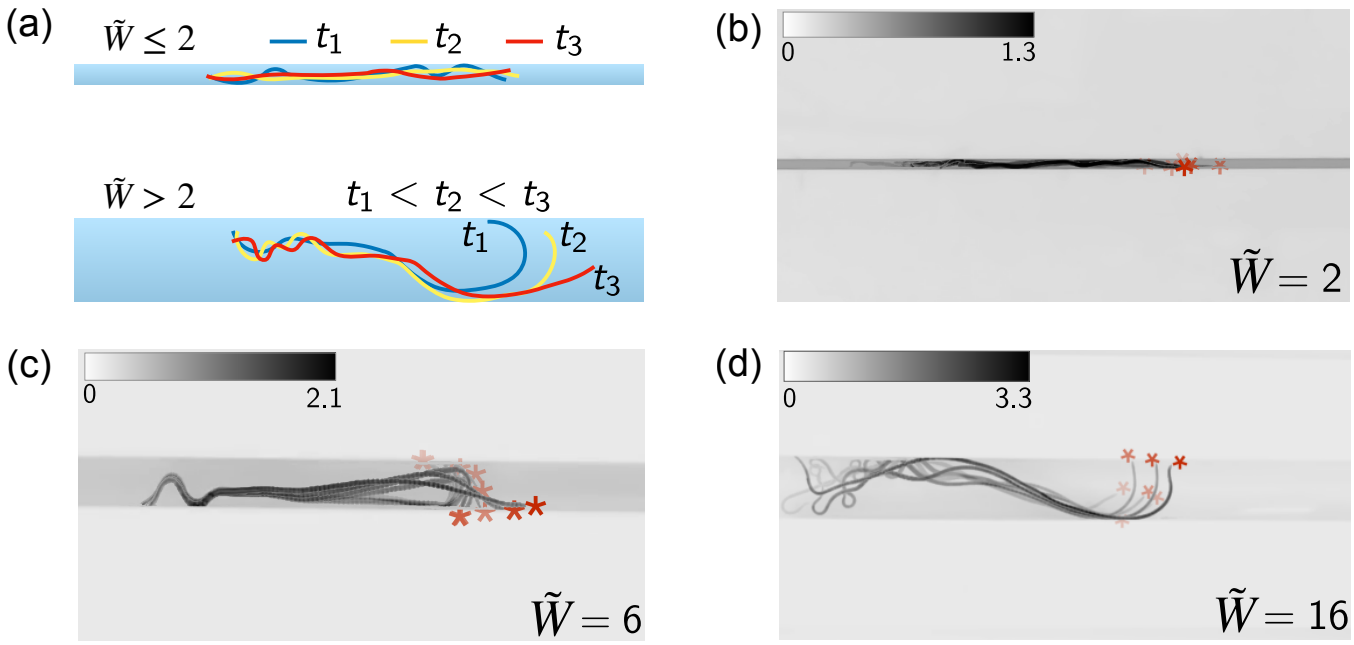


FIG. S3. (a) Schematic of stop-go dynamics of a worm under different confinement strengths. t_1, t_2, and t_3 denote three representative time points used to illustrate worm configurations during a pause. (b)-(c) Overlaid images of the worm during pauses in various confinement, illustrating the associated conformational changes. The head of the worm, marked in red, explores locally while the center of mass of the worm remains unchanged. The gray colormap denotes time in seconds, with lighter shades indicating earlier times. The corresponding $\tilde{W}$ is indicated in each panel.

traversal time is spent in pause and reorientation events whose timescale is set primarily by the animal's gait and is therefore weakly dependent on $\tilde{W}$ once continuous wall guidance is lost. This produces the observed departure from the intermediate power-law regime and the apparent saturation of $t_{\rm esc}$. Consistently, the worm exhibits an intermediate sub-diffusive plateau in the local MSD exponent (SI Fig. S7b), a feature absent in the AP model. In addition, failed entries become more common at large $\tilde{W}$, where the worm can reorient and exit back through the entry side (SI Movie SM2 at $\tilde{W} = 16$). Since Fig. 1c (main text) reports $t_{\rm esc}$ only for successful translocations, the increasing fraction of failed events at large $\tilde{W}$ further biases the mean $t_{\rm esc}$ toward a plateau.

A similar crossover appears in the AP simulations, although more weakly. Unlike the living worms, the active polymer (AP) model does not exhibit intrinsic stop–go dynamics or extended pauses. The $t_{\rm pause}$ distributions broaden only slightly with width, and the kymographs show only brief vertical segments rather than extended stalls (Fig. S5). Instead, in the AP model the relevant crossover is set by the transverse exploration time $t_{\rm sat}$, defined as the time for the rMSD to reach saturation. $t_{\rm sat}$ increases with width but saturates beyond $\tilde{W} \sim 4$ (SI Fig. S8). Once transverse exploration is no longer the limiting step, further increases in $\tilde{W}$ primarily increase the probability of axial reversals rather than continuously increasing a geometric exploration timescale, leading to the milder deviation from a single power law at the largest simulated widths.

The difference in scaling exponents between experiment and simulation arises from two main contributions.

First, the effective dimensionality of transverse exploration differs between experiments and simulations. The simulations are performed in a fully three-dimensional cylindrical channel, where the accessible transverse phase space grows with the channel cross-sectional area ($\sim W^2$). In contrast, the experiments are conducted in an open-top, quasi-2D channel with a fixed depth set by the acrylic thickness. Out-of-plane motion is strongly suppressed, although limited local head exploration and reorientation can still occur. The effective transverse phase space relevant for reorientation in the experiments therefore grows more nearly linearly with W than quadratically. If the probability of reorientation or reversal scales with the available transverse phase space, this reduction by one effective transverse dimension can lead to a smaller apparent width exponent in the experiments.

Second, living worms possess an intrinsic stop-go timescale associated with gait variability in confinement and intermittent pauses, which is absent in the AP model, where propulsion is constant. Schematically, we may write $t_{\rm esc} \simeq t_{\rm th} + t_{\rm wait}$, where $t_{\rm th}$ is the ideal forward traversal time, set by the propulsion speed and filament length and, $t_{\rm wait}$ captures delays in forward motion, eg. stop-go dynamics, re-orientation events, and conformational fluctuation.

In the experiments, $t_{\rm wait}$ contains both the pause time associated with the stop-go motion, $t_{\rm pause}$, and the time associated with reorientation events. In contrast, in the AP simulations, $t_{\rm wait}$ is governed primarily by reorientation dynamics, since propulsion is constant and no intrinsic stop-go timescale is present. Over the limited range of channel widths studied here, fitting the total escape time to a single power-law therefore yields a smaller effective exponent in the experiments than in the reorientation-dominated AP simulations.

This interpretation is supported by our MSD analysis, which shows an intermediate subdiffusive plateau associated with pauses in the worm dynamics, a feature absent in the AP model. We therefore interpret the worm exponent as an effective exponent obtained over the finite range of channel widths studied here, rather than as a universal exponent expected to match the minimal AP model exactly.

More generally, we can write the escape time as $t_{\rm esc} = t_{\rm th} + t_{\rm wait}$, where $t_{\rm th}$ is the forward axial motion of the filament that depends on propulsion speed and length of the filament. $t_{\rm wait}$ captures all delays in forward motion, eg. stop-go dynamics, re-orientation events, conformational fluctuation. In experiments, $t_{\rm wait}$ is associated to the stop-go motion of the worm, $t_{\rm pause}$ and the time scale associated with the re-orientation. In contrast, In AP simulations, $t_{\rm wait}$ is almost governed by the re-orientation dynamics, as the propulsion is constant and no intrinsic stop-go dynamics is present. Over a limited range of channel widths, a single power-law fit to the experimental escape time yields a smaller effective exponent than in the reorientation-dominated AP simulations.

We additionally analyzed the pause-time statistics of freely moving worms using their center-of-mass motion. Even in the absence of confinement, free worms exhibit intermittent exploratory behavior characterized by alternating phases of forward motion, slowing, reorientation, and occasional long pauses. The pause-time distributions display broader-than-exponential tails, suggesting that free worm locomotion already contains intermittent exploratory fluctuations prior to confinement (see Fig. S6). This indicates that intermittent, multi-timescale pausing is an inherent feature of *Lumbriculus variegatus* locomotion, present before any confinement is imposed.

VI. TIME EVOLUTION OF THE MSD EXPONENT ($\beta(t)$)

To characterize the transient dynamics of the filament in confinement, we compute the time-dependent MSD exponent $\beta(t) = \frac{\partial \ln(\mathrm{MSD}(t))}{\partial \ln(t)}$, as shown in Fig. S7. Under strong confinement or for stiff polymers, $\beta \sim 1.9$, throughout the translocation, consistent with nearly ballistic, directed motion (Fig. S7a). In contrast, when $\tilde{W} > 2$ and polymers are flexible, $\beta(t)$ becomes non-monotonic: it rises to $\gtrsim 1.8$ at intermediate times, then crosses over to $\beta \sim 1.1$ as reorientations decorrelate directional persistence.

For worms, β follows an overall rising trend, exhibiting a long-term superdiffusive behavior ($\beta \sim 1.8$) consistent with simulations, with a short term intermediate subdiffusive plateau $\beta \in [0, 1.5]$ arising from intermittent pauses/reorientations; this plateau is absent in the AP model due to constant propulsion (Fig. S7b).

VII. SATURATION TIME OF RADIAL MSD

We quantify transverse exploration by the radial mean-squared displacement $\mathrm{rMSD} = \langle \frac{1}{N} \sum_{i=1}^{N} |\mathbf{r}_i^{\perp}(t+\tau) - \mathbf{r}_i^{\perp}(\tau)|^2 \rangle$, with $\mathbf{r}_i^{\perp}(t) = \{y_i(t), z_i(t)\}$ (main text Fig.2c). The rMSD grows and then saturates at $\mathrm{rMSD}_{\rm sat}$. $\mathrm{rMSD}_{\rm sat}$ increases with channel width, indicating enhanced activity-driven radial exploration of monomers and is largely independent of filament stiffness. We find that $\mathrm{rMSD} \sim \tilde{W}^{2.5}$, steeper than the W^2 scaling expected for a passive tracer. The super-quadratic growth reflects persistent self-propulsion and wall-following (boundary dwelling and sliding), which increase the fraction of the contour near the walls and produce larger cross-sectional excursions than geometry alone would predict.

The time to reach its saturation value, referred to as the saturation time $t_{\rm sat}$ also increases with $\tilde{W}$ and then saturates beyond $\tilde{W} > 3.0$, Fig. S8 indicating confinement-limited dynamics. Notably, $t_{\rm sat}$ shows no dependence on stiffness (data collapse across $\tilde{\ell_{\mathsf{p}}}$), indicating geometry-confinement and self-propulsion, not bending fluctuations, set the timescale. Although $t_{\rm sat} << t_{\rm th}$, it becomes relevant in the stiff limit.

VIII. END TO END VECTOR ANALYSIS AND REORIENTATION DYNAMICS

We analyze the end-to-end vector $\mathbf{R}_E(t) = \mathbf{R}_N(t) - \mathbf{R}_1(t)$ to probe orientation, where $\mathbf{R}_1(t)$, $\mathbf{R}_N(t)$ are coordinates of the first and last monomers of the polymer respectively. The autocorrelation of the end-to-end vector for active filaments under various confinements is shown in Fig S9(a), computed over the interval when the entire polymer is inside the channel. In strong confinement ($\tilde{W} \leq 2.0$), the end-to-end vector remains close to 1, and the correlation

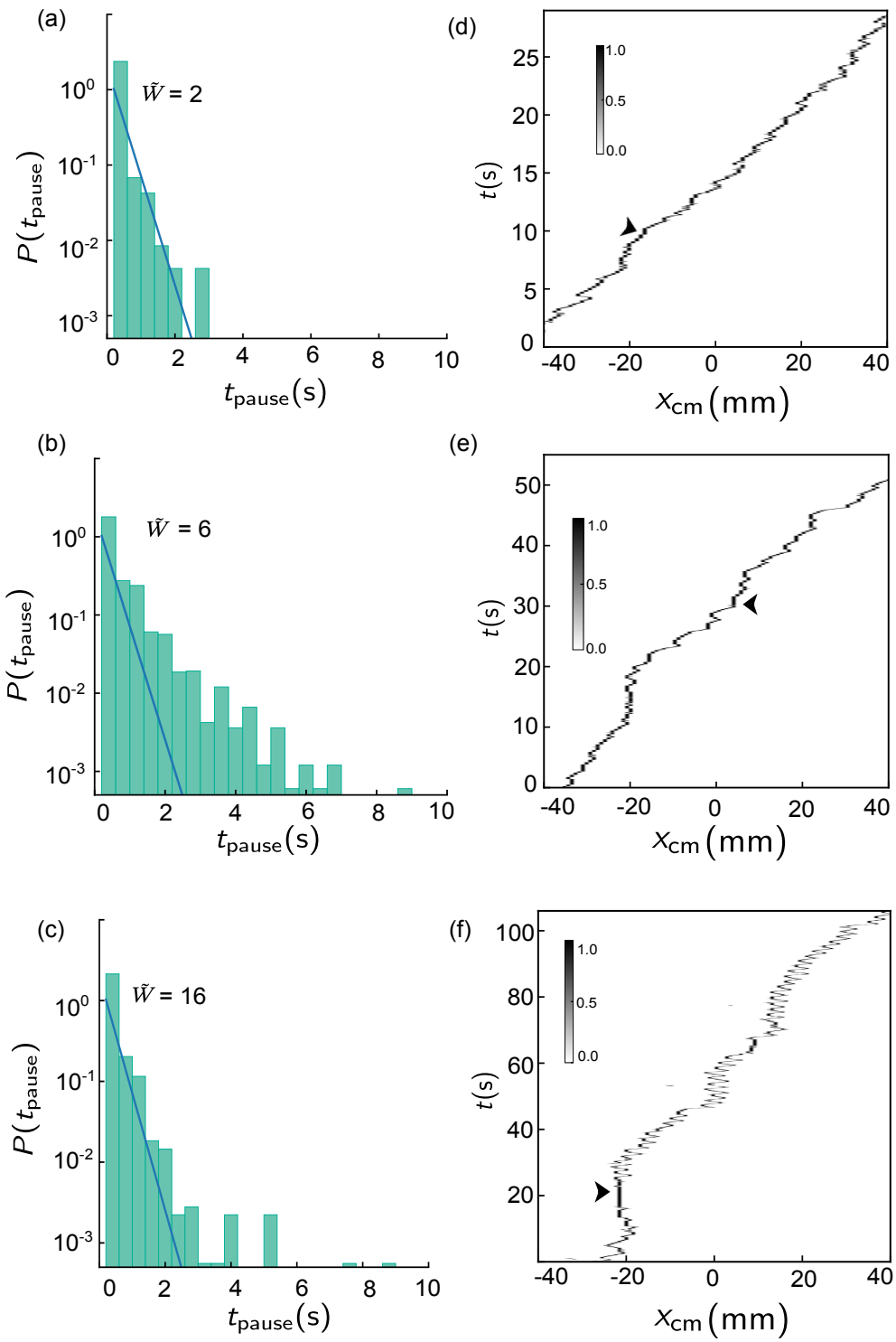


FIG. S4. (a)-(c) Pause-duration statistics of the worm under various confinements. Probability densities $P(t_{\text{pause}})$ of pause durations t_{pause} for worms at $\tilde{W} = 2, 6, 16$. As $\tilde{W}$ increases, $P(t_{\text{pause}})$ broadens and develops a pronounced long-time tail, indicating the emergence of rare but extended pauses. The straight lines are guide-to-the-eye exponential fits. (d)–(f) Corresponding single-trajectory kymographs of the center-of-mass position along the channel axis, x_{cm}, illustrating intermittent stop–go dynamics. The arrows mark long pauses.

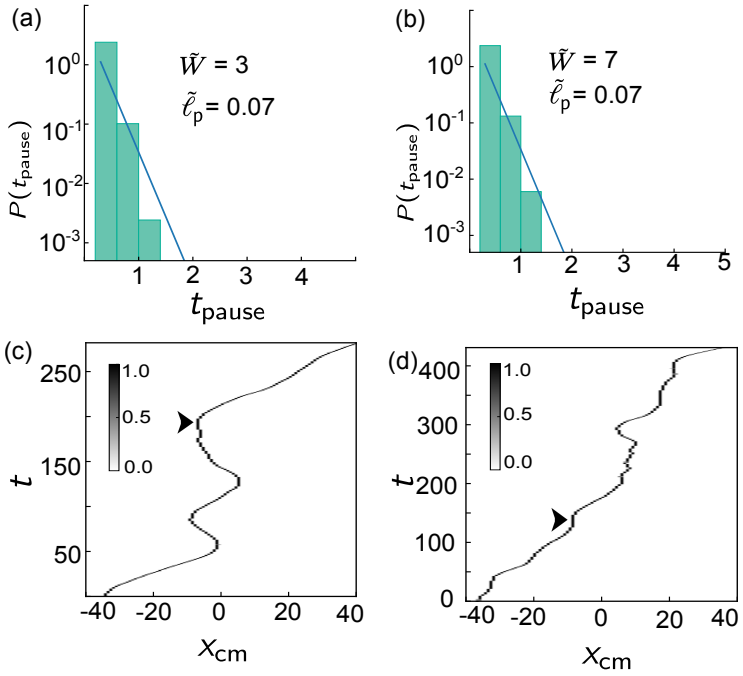


FIG. S5. Pause-duration statistics and kymographs of confined active polymers (AP). (a)-(b) In contrast to the experiments, the simulations exhibit predominantly short pauses with approximately exponential $P(t_{\rm pause})$. (c)-(d) Representative single-trajectory kymographs of the center-of-mass position along the channel axis, $x_{\rm cm}$, showing short pauses indicated by the black arrow. The values of $\tilde{W}$ and $\tilde{\ell}_{\rm p}$ are indicated in each panel, and time is measured in units of τ.

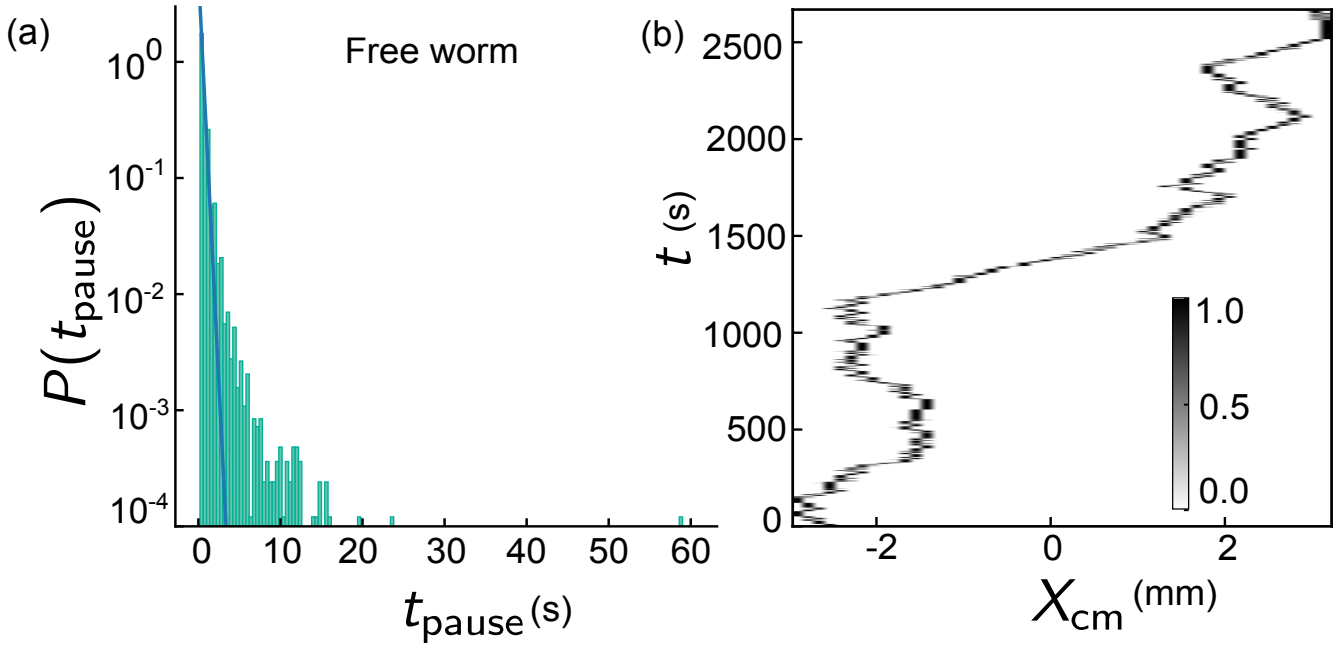


FIG. S6. (a) Pause-time probability distribution $P(t_{\rm pause})$ for a freely moving worm obtained from the center-of-mass dynamics. The distribution exhibits intermittent exploratory dynamics, with repeated slowing, reorientation, and occasional long pauses. These long pauses may arise when the worm undergoes self-contact, transient knotting, body-wall interactions that temporarily hinder forward motion. As a result, the long-time tail deviates from a simple exponential decay, suggesting that pause events are not governed by a single characteristic timescale. (b) Kymograph showing the temporal evolution of the worm center-of-mass position X_{cm} as a function of time, showing intermittent motion.

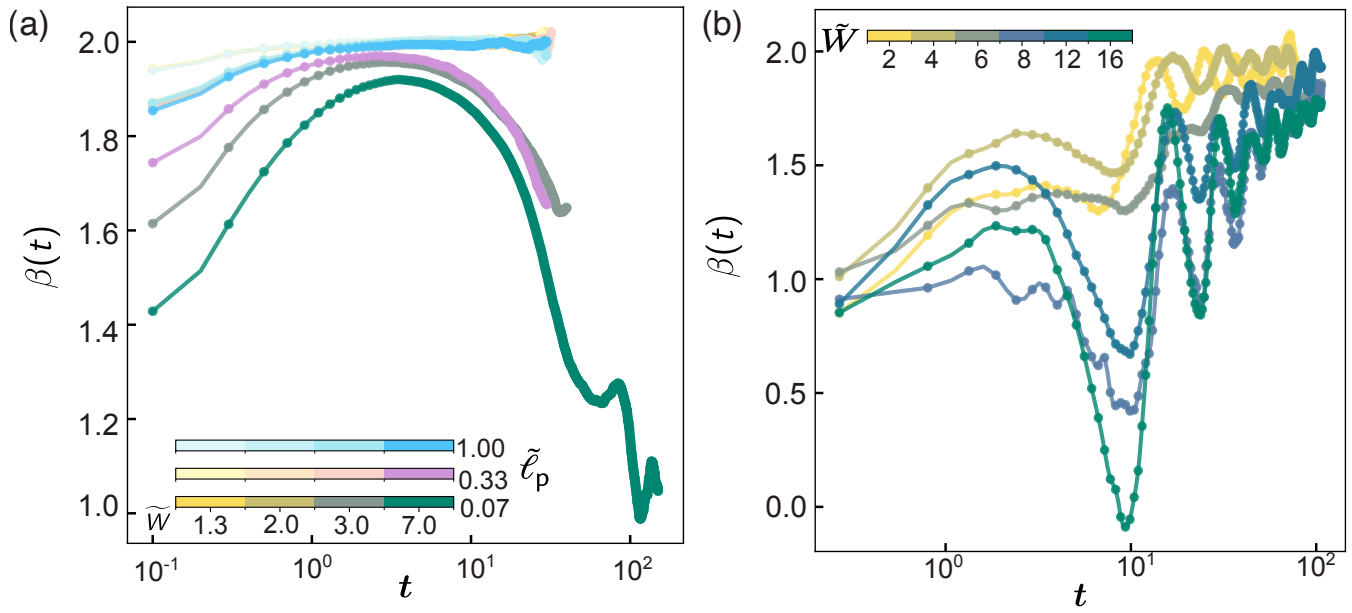


FIG. S7. **Local MSD exponent** $\beta(t)$**.** (a) From AP simulations for different values of $\tilde{\ell}_p$ and $\tilde{W}$ (b) for worms in different confinement.

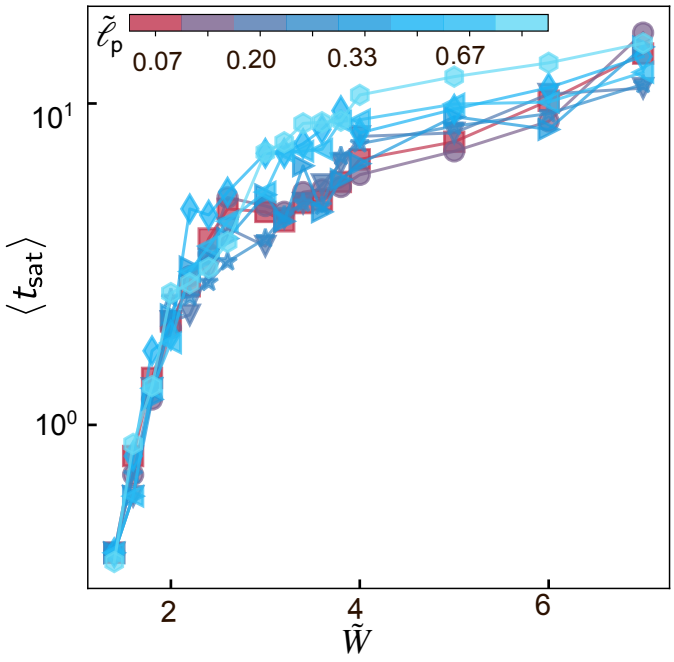


FIG. S8. The saturation time ($t_{\rm sat}$) of rMSD as a function of channel width; initially increases with $\tilde{W}$ and then saturates, showing no dependence on filament stiffness.

shows negligible decay over time. When $\tilde{W} > 2.0$, the correlation decays in a compressed exponential manner, reflecting abrupt reorientations due to persistent propulsion and confinement-induced re-alignment.

To quantify the reorientation of the polymer within the confining channel, we measure the alignment of $\mathbf{R}_E(t)$, with the channel axis $\hat{\mathbf{x}}$, i.e., $sgn[\hat{\mathbf{R}}_E(t) \cdot \hat{\mathbf{x}}]$, Fig S9(b). Each sign change indicates a reorientation of $\mathbf{R_E}$ relative to $\hat{\mathbf{x}}$, and the total number of such sign changes over the translocation trajectory provides a direct number of the axial reversals of the filament (main text Fig 2(d)).

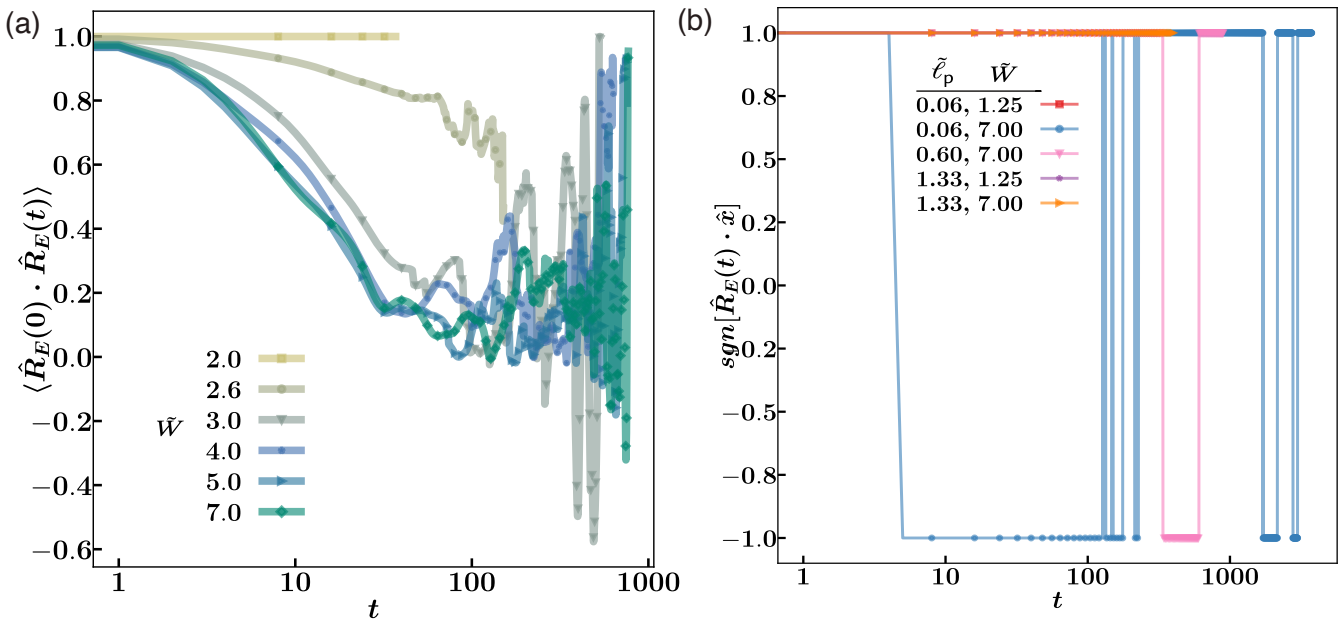


FIG. S9. **Orientation dynamics in confinement.** (a) Autocorrelation of end-to-end vector $\mathbf{R}_E(t) \cdot \mathbf{R}_E(0)$: near-constant under strong confinement ($\tilde{W} \leq 2.0$); and compressed-exponential-like decay in wider channel due to re-orientations. (b) Axial alignment signal $sgn[\hat{\mathbf{R}}_E(t) \cdot \hat{\mathbf{x}}]$: step-like toggles mark reorientation events used to compute reorientation counts N_{RE}.

IX. EFFECT OF PÉCLET NUMBER ON TRANSLOCATION DYNAMICS

To examine whether activity introduces an additional control length scale, we performed simulations by varying f_p, corresponding to the Péclet number, $Pe = 5$ and $Pe = 20$, in addition to $Pe = 10$ reported in the main manuscript. As shown in Fig. S10 (a)-(b), increasing activity systematically reduces the escape time. When t_{esc} is normalized by the axial traversal time t_{th}, the data for different Pe collapse for each stiffness Fig.S10(c), indicating that activity primarily rescales the overall transport timescale while the confinement-stiffness dependence remains unchanged. Consistent with this, the inset of Fig.S10(c) shows approximately $t_{\mathrm{esc}} \propto 1/Pe$.

To test whether a Pe-dependent scaling variable can be constructed, we define an effective run length ℓ_{a} as the ensemble-averaged axial distance traveled by the center of mass between successive axial reversal events. In an unconfined active filament, self-propulsion produces persistent motion, while stochastic rotational fluctuations gradually randomize the filament orientation, thereby limiting the persistence length of the trajectory. Under confinement, these reorientation events are modified by interactions between the polymer and the channel walls, which can deflect the filament and alter its orientation. To identify the duration of such events, we monitor the alignment of the end-to-end vector $\mathbf{R}_E(t)$ with the channel axis, $\hat{x}$, i.e., $sgn[\mathbf{R}_E(t) \cdot \hat{x}]$. Each sign change corresponds to a reorientation of the filament relative to the channel axis, and the total number of such sign changes along the trajectory provides the number of axial reversals (N_{RE}). The effective run length is then computed as the ensemble-averaged axial distance traveled by the center of mass during intervals between successive reorientation events. We define the effective run length as,

$$\ell_{\mathsf{a}} = \left\langle \frac{1}{N_{\mathrm{run}}} \sum_{i=1}^{N_{\mathrm{run}}} \ell_{\mathsf{a}}^{(i)} \right\rangle = \left\langle \frac{1}{N_{\mathrm{run}}} \sum_{i=1}^{N_{\mathrm{run}}} \sum_{j=t_i}^{t_i+N_t^{(i)}-1} |x_{\mathrm{cm}}(t_{j+1}) - x_{\mathrm{cm}}(t_j)| \right\rangle, \tag{7}$$

where N_{run} is the number of runs of fixed polarity in a successful translocation trajectory ($N_{\mathrm{run}} = N_{\mathrm{RE}} + 1$). The quantity ℓ_a^i is the axial distance traveled by the polymer center of mass during the i-th run. The angular brackets $\langle \cdot \rangle$ denote an ensemble average over successful translocation trajectories. The time t_i marks the beginning of the i-th run, and $N_t^{(i)}$ is its duration. The center of mass position of the polymer is x_{cm}. t_j and t_{j+1} are consecutive time steps within the i-th run.

Fig. S11(a) shows that the effective run length depends on the channel width and increases slightly with activity for flexible filaments, reflecting longer persistent motion of the polymer at larger Pe. We examined the mean number of reorientation events as a function of $\tilde{W}^2/\tilde{\ell}_{\mathsf{a}}$ in Fig. S11(b) for $Pe = 10$. $\langle N_{\mathrm{RE}} \rangle$ increases with this ratio, indicating

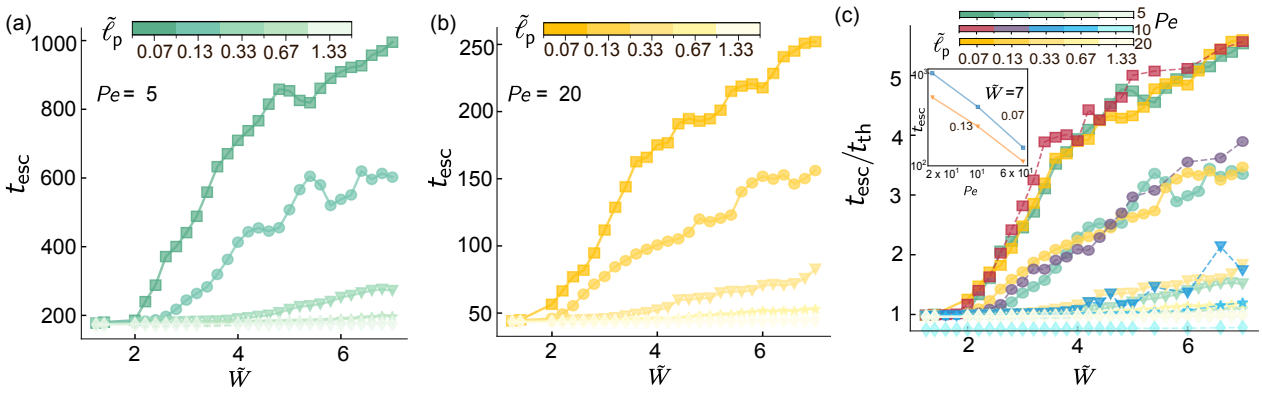


FIG. S10. Translocation time as a function of scaled width for polymers of different stiffnesses. (a)$Pe = 5$. (b)$Pe = 20$. (c) Escape time normalized by the axial traversal time $t_{\rm th}$ for $Pe = 5, 10, 20$. The collapse of the data for each stiffness indicates that activity rescales the axial transport time without altering the confinement-stiffness dependence. Inset: $t_{\rm esc}$ versus Péclet number at fixed $\tilde{W} = 7$, showing approximately $t_{\rm esc} \propto Pe^{-1}$.

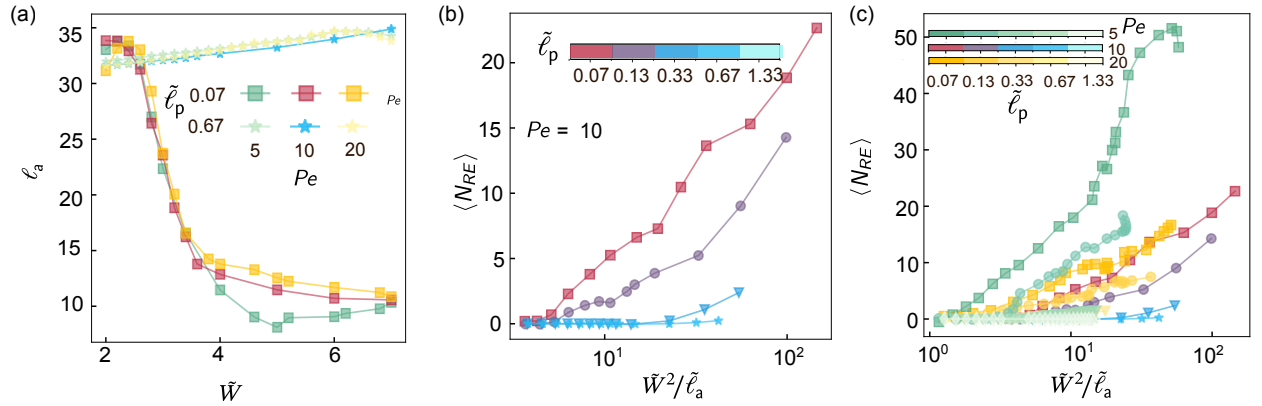


FIG. S11. (a) Effective run length (ℓ_a) as a function of scaled width $\tilde{W}$ for different stiffness and Péclet numbers. ℓ_a decreases with increasing width and increases modestly with activity for flexible filaments. (b) $\langle N_{\rm RE} \rangle$ plotted against $\tilde{W}^2/\tilde{\ell}_a$ for $Pe = 10$, where $\tilde{\ell}_a = \ell_a/L$. Although $\langle N_{\rm RE} \rangle$ decreases as the stiffness is increased and no data collapse is observed across different stiffnesses. (c) $\langle N_{\rm RE} \rangle$ versus $\tilde{W}^2/\tilde{\ell}_a$ for $Pe = 5, 10, 20$. Although the curves follow similar trends, they do not collapse across activity, indicating that activity modifies the timescale of motion rather than the underlying reorientation mechanism.

that larger transverse exploration relative to the run length leads to more frequent reorientations. However, the curves corresponding to different stiffness values do not collapse onto a single master curve, suggesting that filament flexibility remains an additional control parameter in the reorientation dynamics. In Fig. S11(c) we compare different activities. While increasing activity generally reduces the number of reorientation events, the curves corresponding to different Pe do not collapse when plotted against $\tilde{W}^2/\tilde{\ell}_a$. This indicates that although activity slightly modifies the effective run length ℓ_a, the confinement and filament stiffness determine the overall translocation dynamics through $\tilde{W}^2/\tilde{\ell}_a$, while activity primarily rescales the time scale of motion without altering the underlying reorientation mechanism.

X. ROLE OF POLYMER STIFFNESS IN CAVITY EXPLORATION AND TRANSLOCATION

The complete transport process of the worm in the experimental setup involves two stages: (i) escape from or capture into the circular chamber, characterized by a cavity timescale $t_{\rm cavity}$, and (ii) translocation through the connecting channel, characterized by the escape time $t_{\rm esc}$. A previous numerical study of an active filament in a 3D spherical cavity [8] demonstrated that increasing bending rigidity reduces the $t_{\rm cavity}$, because stiffer filaments maintain more persistent motion and undergo fewer reorientation events, whereas flexible filaments reorient more

frequently and spend longer searching within the cavity. That study also showed that the cavity radius influences the corresponding search time. In the present work, we find that increasing filament stiffness similarly reduces the channel translocation time $t_{\rm esc}$, due to fewer reorientation events inside the channel. These observations suggest that filament stiffness can affect both stages of the full pore-to-pore transport process. However, in a quasi-two-dimensional chamber-channel geometry such as the one studied here, a very stiff rod-like filament may also experience delayed cavity escape because pore entry requires more precise alignment at the channel entrance. Determining whether these competing effects produce an optimal intermediate stiffness in the total transport time would require simulations that explicitly couple cavity exploration and channel translocation, which is a natural extension of this present work.

XI. EFFECTIVE PERSISTENCE LENGTH OF ACTIVE POLYMER IN CONFINEMENT

We computed the tangent–tangent correlation function, $C(s) = \langle \mathbf{t}_i \cdot \mathbf{t}_{i+s} \rangle$, where $(\mathbf{t}_i)$ is the unit tangent vector along the polymer contour and (s) is the contour distance along the polymer backbone (see Fig S12(a)). The correlations were averaged over contour positions, time frames, and independent successful runs.

We find that, for weak confinement, the correlations decay approximately toward zero, whereas, under strong confinement, the correlations develop a finite plateau induced by confinement and activity. Therefore, we employed the generalized form $C(s) = C_\infty + (1 - C_\infty)\exp(-s/\ell_{\mathsf{a}}^{\rm eff})$, where (C_∞) represents the finite plateau in the tangent–tangent correlation at large contour separations and $(\ell_{\mathsf{a}}^{\rm eff})$ represents an effective persistence length (or orientational correlation length). The resulting $(\ell_{\mathsf{a}}^{\rm eff})$ shows a moderate dependence on (Pe), confirming that activity indeed modifies the filament conformations (see Fig S12(b)). However, the dominant trends depend on the confinement and filament stiffness. In particular, confinement suppresses bending fluctuations and promotes rod-like alignment, leading to enhanced orientational correlations. These results support our interpretation that activity primarily modifies the effective orientational persistence of the filament, while confinement geometry continues to govern the overall reorientation dynamics.

It is clear that the effective persistence length $\ell_{\mathsf{a}}^{\rm eff}$, extracted from tangent correlations, depends on confinement width, activity, and filament stiffness, indicating that it is not an independent material parameter but already incorporates confinement, and activity-induced conformational changes. Similarly, plotting $\langle N_{\rm RE} \rangle$ against $\tilde{W}^2/\tilde{\ell}_{\mathsf{a}}$ does not produce the collapse of the data (see Fig S11(c)). In contrast, as shown in the main text Fig 2(d) and Fig S13, plotting $\langle N_{\rm RE} \rangle$ as a function of $\tilde{W}^2/\tilde{\ell}_{\mathsf{a}}$ yields an approximate collapse of data across different activity strengths, suggesting that the reorientation dynamics are primarily governed by the interplay between transverse confinement and the intrinsic bending stiffness of the filament. The observed scaling therefore indicates that the ratio, $\tilde{W}^2/\tilde{\ell}_{\mathsf{a}}$ provides a much simpler and more robust description of the confinement-controlled reorientation dynamics in the present system.

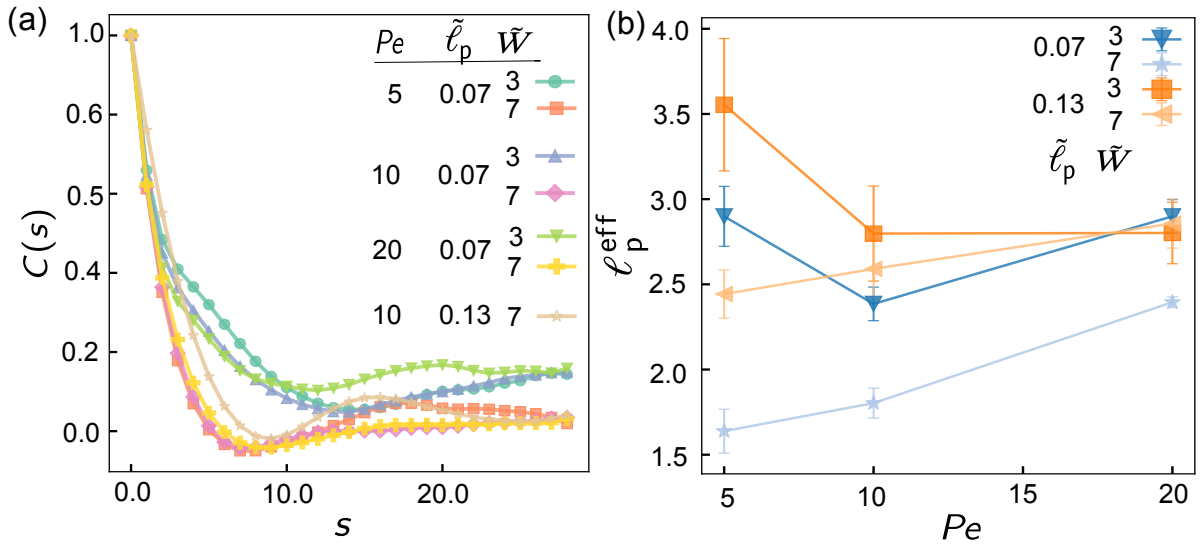


FIG. S12. (a) Tangent–tangent correlation function $C(s)$ under different confinement widths $\tilde{W}$ (3 and 7), stiffness $\tilde{\ell}_p$ (0.07 and 0.13) and activity strengths Pe (5, 10, and 20). (b) The effective stiffness ℓ_p^{eff} extracted from the $C(s)$ as a function of Pe. In wider channel, ℓ_p^{eff} exhibits an increasing tendency with activity, reflecting enhanced orientational persistence of the active filament. However, under stronger confinement ℓ_p^{eff} shows a weaker dependence on Pe, consistent with confinement-induced suppression of transverse fluctuations.

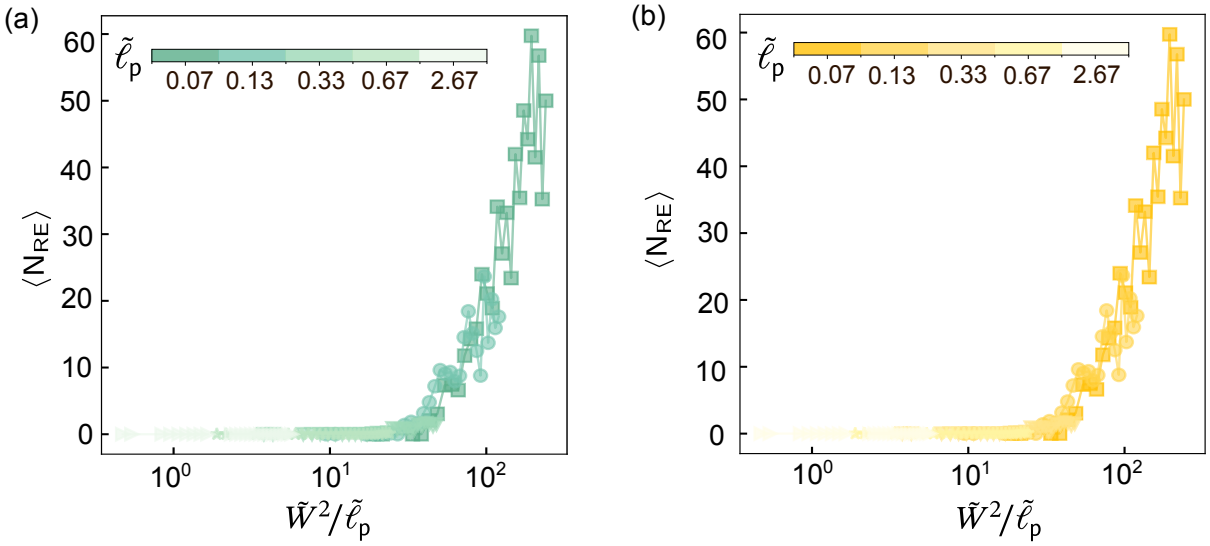


FIG. S13. Average number of re-orientations as a function of the scaled confinement parameter $\tilde{W}^2/\tilde{\ell}_p$ for different Péclet number (a)$Pe = 5$ (b) $Pe = 20$. The data exhibit an approximate collapse onto a single curve, suggesting that $\tilde{W}^2/\tilde{\ell}_p$ captures the dominant scaling governing reorientation dynamics.

XII. SI MOVIES

Notes. Simulation movies were rendered from frames recorded every $0.1\,\tau$; playback is accelerated $8\times$ for visualization. In the figure captions and legends, $\tilde{\ell}_\mathsf{p}$ and $\tilde{W}$ are rounded to two decimal places for clarity.

1. **SM1.** Comparative translocation of blackworms (*Lumbriculus variegatus*) across confinements ($\tilde{W} = 2, 4, 12$). Strong confinement yields faster escape.
2. **SM2.** Failed translocation (worm). A worm in a wide channel ($\tilde{W} = 16$) reorients and exits from the entry side, illustrating a failed event.
3. **SM3.** Successful translocation with reorientation (worm). A worm in a wide channel ($\tilde{W} = 16$) undergoes body reorientation during a successful escape.
4. **SM4.** Strong confinement (active polymer). Flexible polymer with $\tilde{\ell}_\mathsf{p} = 0.067$ at $\tilde{W} = 1.245$. All beads propel along the local tangent; no axial reorientations are observed.
5. **SM5.** Strong confinement (active polymer). Stiffer polymer with $\tilde{\ell}_\mathsf{p} = 1.33$ at $\tilde{W} = 1.245$. Dynamics closely match the flexible case, indicating weak stiffness dependence in this regime.
6. **SM6.** Wide channel, de Gennes–like regime (active polymer). Flexible polymer with $\tilde{\ell}_\mathsf{p} = 0.067$ at $\tilde{W} = 7.0$ exhibits frequent reorientations and shape changes.
7. **SM7.** Stiffer polymer with $\tilde{\ell}_\mathsf{p} = 1.33$ at $\tilde{W} = 7.0$ maintains an extended, axis-aligned, Odijk-like conformation with short wall deflections.
8. **SM8.** Failed translocation (active polymer). Flexible polymer with $\tilde{\ell}_\mathsf{p} = 0.067$ at $\tilde{W} = 3.0$ repeatedly reorients within the channel and fails to exit the far end.

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